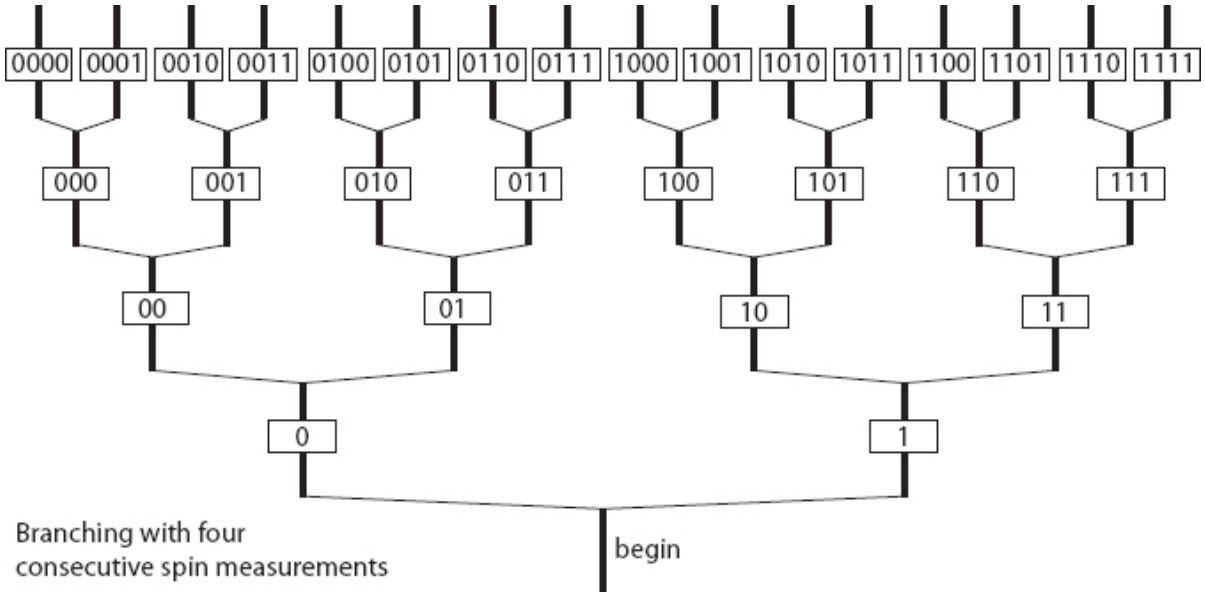


Branching with four consecutive spin measurements

If all of the copies of me in all of those different worlds stuck with the plan of including the obtained number into the text of this book, that means there are over a quadrillion different textual variations of *Something Deeply Hidden* out there in the wave function of the universe. For the most part the variations will be minor, just rearranging some 0's and 1's. But some of those poor versions of me were the unlucky ones who got all 0's or all 1's. What are they thinking right now? Probably they thought the random-number generator was broken. They certainly didn't write precisely the text I am typing at this moment.

Whatever I or the other copies of me might think about this situation, it's quite different from the frequentist paradigm for probabilities. It doesn't make too much sense to talk about the frequency in the limit of an infinite number of trials when every trial returns every result, just somewhere else in the wave function. We need to turn to another way of thinking about what probability is supposed to mean.

Fortunately, an alternative approach to probability exists, and long pre-dates quantum mechanics. That's the notion of *epistemic* probability, having to do with what we know rather than some hypothetical infinite number of trials.

Consider the question "What is the probability that the Philadelphia 76ers will win the 2020 NBA Championship?" (I put a high value on that